

(1c)  $\lfloor x \rfloor = n$  if and only if  $x-1 < n \leq x$ .

Solution: Let,  $\lfloor x \rfloor = n$ . By (1a),  $n \leq x < n+1$ .

$$x < n+1 \Rightarrow x-1 < (n+1)-1 \Rightarrow x-1 < n.$$

$$\therefore x-1 < n \leq x$$

$$\text{Let, } x-1 < n \leq x. \quad (x-1) < n \Rightarrow (x-1)+1 < n+1 \Rightarrow x < n+1$$

$$\therefore n \leq x < n+1 \quad \text{By (1a), } \lfloor x \rfloor = n \quad \square$$

(1d)  $\lceil x \rceil = n$  if and only if  $x \leq n < x+1$

Solution: Let,  $\lceil x \rceil = n$ . By (1b),  $n-1 < x \leq n$ .

$$n-1 < x \Rightarrow (n-1)+1 < x+1 \Rightarrow n < x+1. \quad \therefore x \leq n < x+1$$

$$\text{Let, } x \leq n < x+1. \quad n < x+1 \Rightarrow n-1 < (x+1)-1 \Rightarrow n-1 < x$$

$$\therefore n-1 < x \leq n \quad \text{By (1b), } \lceil x \rceil = n \quad \square$$

$$(2) x-1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x+1$$

Proof: Let,  $\lfloor x \rfloor = p$ , where  $p \in \mathbb{Z}$  from Defn. 12

$$x-1 < \lfloor x \rfloor \leq x \quad (\text{By 1(c)})$$

Let,  $\lceil x \rceil = q$ , where  $q \in \mathbb{Z}$  from Defn. 12

$$x \leq \lceil x \rceil < x+1 \quad (\text{By 1(d)})$$

$$\therefore x-1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x+1 \quad \square$$

$$(3a) \lfloor -x \rfloor = -\lceil x \rceil$$

Proof: Let,  $\lfloor -x \rfloor = p$ . By Defn. 12,  $p \in \mathbb{Z}$ .

$$\text{By (1a), } p \leq -x < p+1. \Rightarrow (-1) \cdot p \geq (-1) \cdot (-x) > (-1) \cdot (p+1)$$

$$\Rightarrow \boxed{-p \geq x > -p-1}.$$

There are no integers in between  $-p$  and  $-p-1$  since,  $-p - (-p-1) = 1$  is the smallest absolute difference between any two arbitrary different integers.  $\therefore -p$  is the smallest integer which is larger than  $x$ . By Defn. 12,  $\lceil x \rceil = -p \Rightarrow -\lceil x \rceil = p$

$$\Rightarrow \boxed{\lfloor -x \rfloor = -\lceil x \rceil} \quad \square$$